

Chapter 1

Functions and Transforms

The goal of this text is to provide the reader with an explanation of the fast Fourier and wavelet transforms. The goal in this chapter is to provide the reader with an explanation of the general idea behind transforms and to introduce some necessary terminology. The topics in this chapter include transforms, fields, and signal classification. The fast Fourier and wavelet transforms are covered in more detail later in the book.

Chapter Goals: After completing this chapter, you should be able to do the following:

- Define *transform*.
- Determine if a given signal is a power or energy signal.
- For a given discrete-time signal of length 4, calculate both the discrete-time Fourier series and the Haar transforms.

1.1. Wavelet Transform

There are two ways to determine the wavelet transform. Most transforms have only one way, a formula. For example, the Laplace transform of a time function $v(t)$ is calculated by the formula

$$V(s) = \int_{-\infty}^{\infty} v(t)e^{-st} dt$$

There is a similar formula for the wavelet transform, but we want to show you the other view in this introduction. This second view is the way the wavelet transform is calculated in practice.

Start with samples of a continuous-time signal over some interval. Let us say that the interval is long enough to include 1024 samples. Supply this discrete-time signal to two digital filters in parallel. Figure 1.1 shows the process, where H_0 is a low-pass filter and H_1 is a high-pass filter. The input is $s(n)$, and $c(n)$ and $d(n)$ are the output terms. The $(\downarrow 2)$ symbol represents downsampling by 2, which means that we throw away every other sample. We keep the 1st, 3rd, 5th, ... samples and throw away the 2nd, 4th, 6th, ... samples. In this scheme $c(n)$ contains 512 samples, as does $d(n)$.

The 512 $d(n)$ terms represent part of the transform (in fact, half of the transform). The other half is extracted from $c(n)$ by repeating the filter-

downsample operation. Figure 1.2 shows exactly what this means. Note that the first filter-downsample terms are c_9 and d_9 . This is because they contain 512 samples and $2^9 = 512$. Similarly, the second-stage output is labeled c_8 and d_8 because it contains 256 terms. The 256 d_8 terms represent an additional one-fourth of the transform. Now you can guess how to calculate the additional terms. Repeating the filter-downsample operation once again on the c_8 terms generates c_7 and d_7 , each containing 128 terms.

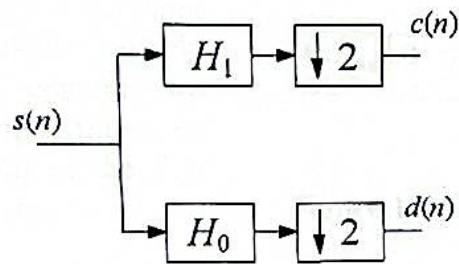


Figure 1.1. The first stage of the transform.

The wavelet transform contains the same number of terms as the signal, 1024 in this example. The terms in the transform consist of d_9 (512 samples), d_8 (256 samples), d_7 (128 samples), d_6 (64 samples), d_5 (32 samples), d_4 (16 samples), d_3 (8 samples), d_2 (4 samples), d_1 (2 samples), d_0 (1 sample), plus the one term in c_0 .

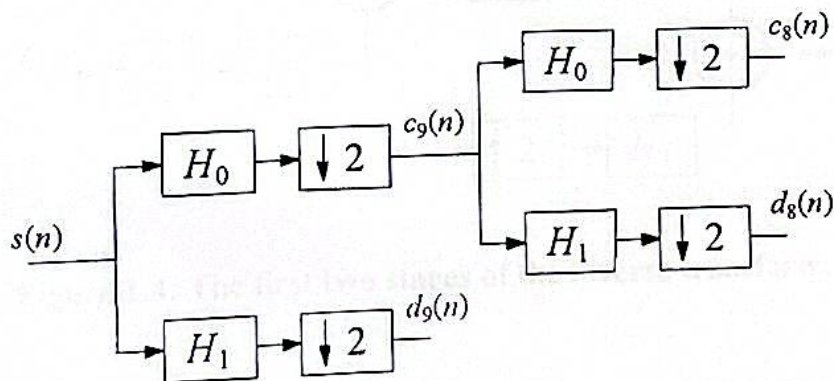


Figure 1.2. The first two stages of the transform.

To find the inverse transform, reverse this process. Start with the two terms c_0 and d_0 . Supply these to two digital filters in parallel, as shown in Fig. 1.3. The ($\uparrow 2$) symbol represents upsampling by 2, which means that we insert zeros between samples. Starting with a signal of length 4 given by $\{1, 2, -1, 3\}$, the resulting upsampled signal is $\{1, 0, 2, 0, -1, 0, 3, 0\}$, which has length 8. Initially, there is one sample at the input of each upsample-filter branch, the output will contain two samples. For proper filters H_2 and H_3 the output of the first summer will be $c_1(n)$.

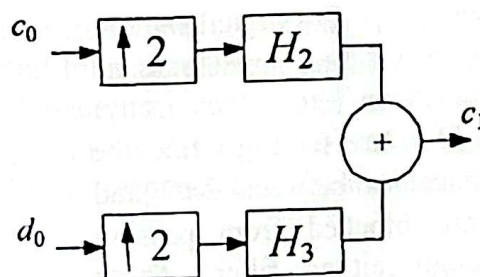


Figure 1.3. The Inverse Transform.

Figure 1.4 shows the first two stages of the inverse transform. This process continues until the last output is $s(n)$, the original signal. Note that we must supply $c_0(n)$, $d_0(n)$, $d_1(n)$, $d_2(n)$, ..., $d_9(n)$ at successive stages of this process. These are the terms that we saved and called the transform in the forward process.

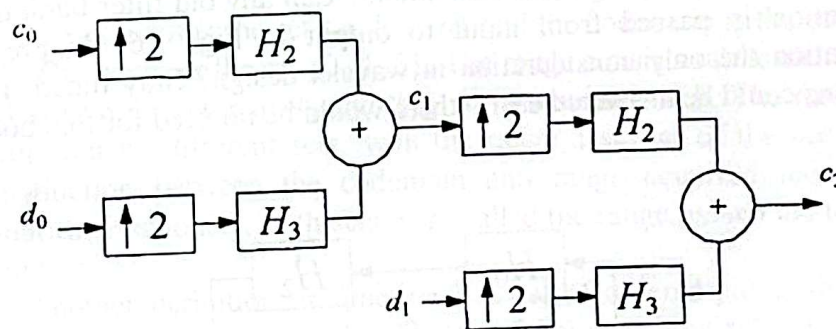


Figure 1.4. The first two stages of the inverse transform.

Little has been said about the filters in this process, except that H_0 is low-pass and H_1 is high-pass. What characteristics must these filters have, and what is their relationship one to another? We can answer some of these

questions now by knowing that $c_0(n)$, $d_0(n)$, $d_1(n)$, $d_2(n)$, ..., $d_9(n)$ must contain the same information as $s(n)$. The remainder of this book should provide enough information to answer most of these questions.

Consider the relationship between H_0 and H_1 . Figure 1.5 shows the situation that will not work. Between them, the two filters must pass all frequencies. For digital signals, this means they must pass all frequencies in the range $0 \leq \Omega < 2\pi$. In Fig. 1.5, the frequencies between $2\pi/3$ and π are blocked from passing through either filter. These filters cannot preserve information.

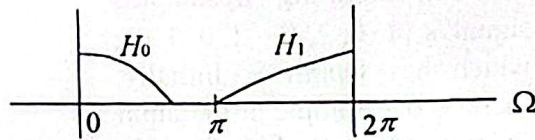


Figure 1.5. Two frequency responses.

Figure 1.6 shows a filter bank that depicts the relationship between the four filters without the $(\downarrow 2)$ and $(\uparrow 2)$ operations. Consider the relationship between H_0 and H_2 , or the top branch in Fig. 1.6. If the output y contains all the information in the input v , then H_2 must also be a low-pass filter. Suppose, to the contrary, that H_0 is low-pass and H_2 is high-pass. Then all frequency components will be blocked from passing through the top branch. Similarly, if H_1 is high-pass, then H_3 must also be high-pass.

This still leaves many questions to be answered. What must be the exact relationship among these four filters? Can any old filter bank do if all information is passed from input to output in Fig. 1.6? Is preserving information the only consideration in wavelet design? Stay tuned. If these questions could be answered easily there would be no need for this book.

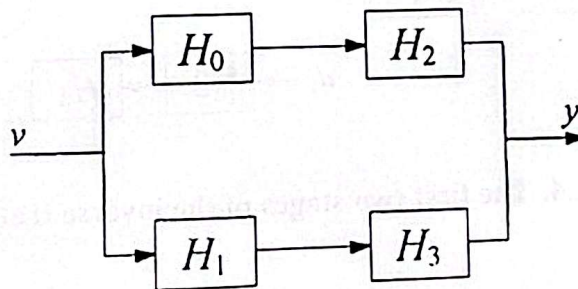


Figure 1.6. Filter bank with all four filters.

1.2. Transforms

A transform is a special type of function. To understand transforms, one must first understand the concept of a function. The reader will recall that a *function* is a relationship between two sets (called the *domain* and *codomain*), and this relationship must satisfy two conditions. First, every element in the domain must correspond to some element in the codomain. Second, no two elements in the codomain can correspond to the same element in the domain. These two requirements can be combined by saying that for each element in the domain there corresponds exactly one element in the codomain.

Figure 1.7 shows a map or function. We write this $f: A \rightarrow B$ and call the function f , the domain A , and the codomain B . Notice that each element in the domain A is matched up with something in B (the first requirement) and that no two elements in B match up with the same element in A (the second requirement.) It is OK for two different elements in A to match with the same element in B .

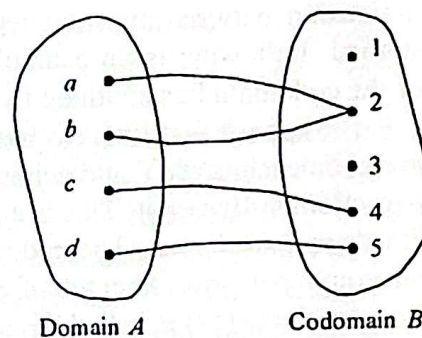


Figure 1.7. The function $f: A \rightarrow B$.

The domain is the set $\{a, b, c, d\}$ and the codomain is the set $\{1, 2, 3, 4, 5\}$. The *range* is the set $\{2, 4, 5\}$. The range consists of the elements in B that have an ancestor in A , and they form a subset of B . The range and codomain can be different sets, with the range a subset of the codomain. This distinction between the codomain and range occurred recently in mathematics. Previously, both sets were called the range, which led to some confusion.

Another definition of function is a set of ordered pairs where the first element is not repeated. Consider the following three sets of ordered pairs.

$(a,2)$	$(a,2)$	$(a,1)$
$(b,2)$	$(b,2)$	$(b,2)$
$(c,4)$	$(a,4)$	$(c,4)$
$(d,5)$	$(d,5)$	$(d,5)$

The first column represents the function f in Fig. 1.7. The second column is not a function because a is repeated. The third column is a function.

Definition 1.1. A function $f: A \rightarrow B$ is a relationship between two sets A and B such that for each element in the first set there corresponds exactly one element in the second set.

Other names for *function* are *map* and *mapping*. In the beginning, the concept of function applied to numbers. The domain and codomain consisted of numbers. These are the functions for which graphs are drawn and formulas are written. Now the concept of function applies to the relationship between any two sets if the two properties for a function are satisfied. Following is an example of a function where neither the domain nor the codomain has anything to do with numbers.

Suppose that I have a basket full of cards, and that an instruction is written on each card. You reach in the basket, select a card, and carry out the instruction written on it. This is a function. The domain is the basket of cards with their instructions. The codomain is the set of all possible actions that you could perform. The set of actions that might actually be performed (those on the cards) is called the *range*, and this is a subset of the codomain.

In modern parlance, the term *range* refers to those elements in the codomain that have an ancestor in the domain. Figure 1.8 shows the domain and codomain as baskets. All values of y fall into the subset of the codomain called the range after being ejected from the box labeled f . The function $y = x^2$ provides a specific example. If the domain and codomain are the set of all real numbers, the range is the set of all non-negative real numbers. Therefore, in this example the range is a proper subset of the codomain. (*Proper* means that some elements in the codomain are not in the range.)

Now for the idea of a transform. The functions of interest to us here also have no numbers in either the domain or codomain. These special functions go by either of two names, *transform* or *operator*. They are functions whose domain and range are themselves sets of functions. If Fig. 1.8 represents a transform, all baskets contain functions instead of numbers. For example, in the Fourier transform the domain contains time functions and the codomain contains frequency functions. The formula for the Fourier transform is given by

$$V(\omega) = \int_{-\infty}^{\infty} v(t) e^{-j\omega t} dt$$

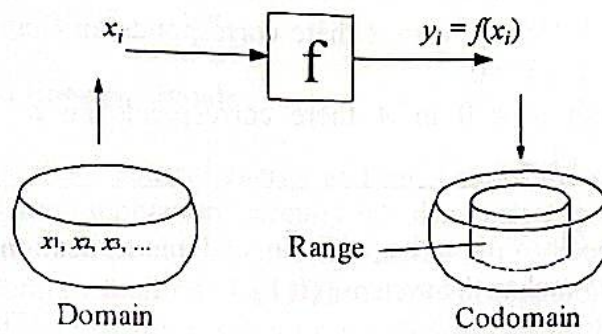


Figure 1.8. The concept of function.

We can think of this in relation to Fig. 1.8 in the following way: Select a time function $v(t)$ and supply it to the black box labeled f . Inside the black box the function $v(t)$ is multiplied by the exponential $e^{-j\omega t}$ and then integrated over all time. The result is the frequency function $V(\omega)$, which pops out of the box and falls into the range. The domain contains all time functions that have a Fourier transform, and the range contains all corresponding frequency functions. The basic idea of a transform is a map where the domain and codomain consist of functions.

Fields

Another necessary concept is that of a *field*. Think of ordinary numbers with the two operations addition and multiplication. This is the model used to define a field. Any set with two operations defined on it that satisfies the following definition is called a field.

Definition 1.2. Let A be a set consisting of the elements $\{a, b, c, \dots\}$. The *field* $(A, +, \cdot)$ is a set together with two operations, $+$ and $\cdot$. These operations, called *addition* and *multiplication*, together satisfy the following conditions:

- (i) $a(b+c) = a \cdot b + a \cdot c$ for all a, b, c in A .
- (ii) $a+(b+c) = (a+b)+c$ for all a, b, c in A .
- (iii) $a(b \cdot c) = (a \cdot b) \cdot c$ for all a, b, c in A .
- (iv) $a+b = b+a$ for all a, b in A .
- (v) $a \cdot b = b \cdot a$ for all a, b in A .
- (vi) There exists an element 0 in A such that $a + 0 = a$ for all a in A .

- (vii) There exists an element 1 in A such that $a \cdot 1 = a$ for all a in A .
- (viii) To each element a in A there corresponds an element $-a$ in A such that $a + (-a) = 0$.
- (ix) To each $a \neq 0$ in A there corresponds an a^{-1} in A such that $a \cdot a^{-1} = 1$.

Example 1.1. Show that the addition and multiplication defined by the following tables makes the system $(\{0,1\}, +, \cdot)$ a field.

$+_2$	0	1
0	0	1
1	1	0

$\cdot_2$	0	1
0	0	0
1	0	1

Solution: This is called the binary field. To show compliance with Definition 1.2, choose values for a , b , and c and determine whether or not properties (i) through (v) are satisfied. For the binary field this gives us 2^3 , or 8, possible combinations. All eight combinations must satisfy these properties. (They do.) Then we must find the 0 element and the 1 element (which are 0 and 1) to satisfy properties (vi) and (vii). The additive inverse for 0 is 0 and for 1 it is 1 [property (viii)]. The multiplicative inverse of 1 is 1 because $1 \cdot 1 = 1$. Thus, the addition and multiplication tables above make the system $(\{0,1\}, +, \cdot)$ a field. ■

Example 1.2. Give the tables $+$ and $\cdot$ to make $(\{0,1,2\}, +, \cdot)$ a field.

Solution:

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot_3$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

The fields used most often in this book are the real number system and the complex number system.

1.3. Power and Energy Signals

A resistor dissipates electrical energy as heat. One of the more memorable experiments in any sophomore circuits lab demonstrates this fact with a candle leaning against a hot resistor. Heat the resistor by impressing a steady voltage of moderate value across it. If the energy dissipated, candle size, and power rating of the resistor are about right, the melting of the candle, which rests against the resistor, will be proportional to the electrical energy dissipated. This experiment demonstrates not only that energy is dissipated as heat, but also that energy is dissipated over time. If the voltage is not too large, the resistor warms to a constant temperature and remains there. In other words, it reaches a steady state. Thus, *average power* is dissipated in the resistor. This implies the following:

1. The heat produced is proportional to average power. Note the emphasis on average power, not instantaneous power.
2. This implies a steady-state signal. Ideally, a steady-state signal exists forever, starting before the world was formed and lasting until long after the world has burned to a crisp. For the lab experiment, however, the signal must last only for the duration of the experiment. Thus, the idealization to infinite time is one of mathematical convenience.

For a continuous-time signal voltage $v(t)$ impressed across a resistor, the average power is defined by

$$P = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a \frac{v^2(t)}{R} dt \quad (1.1)$$

To make this quantity depend only on the signal, set the resistor equal to 1 ohm. The total power, average power, or mean-square value of the signal is defined to be

$$P = \lim_{a \rightarrow \infty} \frac{1}{2a} \int_{-a}^a v^2(t) dt \quad (1.2)$$

The square root of P is the root mean square (rms) value. Notice that current through the resistor could be used just as easily as the voltage in the

definition of power. Our definition of power in a signal applies to any signal, whether it is voltage, current, or some nonelectrical function such as pressure or velocity.

This serves to introduce the classification of signals into one of two categories: energy signals or power signals. A signal $v(t)$ is called a *power signal* if the expression in Eq. 1.2 is finite (i.e., $0 < P < \infty$). A signal $v(t)$ is called an *energy signal* if the energy is finite, where the energy E is given by

$$E = \int_{-\infty}^{\infty} v^2(t) dt \quad (1.3)$$

This is the total energy, not the instantaneous energy. Our concern is with those signals that can be classified into one of the two categories, but you should understand that there are other signals. A ramp signal, or an increasing exponential signal, has infinite power, and therefore is neither a power nor an energy signal. Our needs do not include such signals, so in this book every signal is either an energy signal or a power signal. Notice that any signal that lasts for only a finite duration is an energy signal. A signal, which lasts for an infinite duration can be either a power or an energy signal, depending on the value of the expressions in Eqs. 1.2 and 1.3.

Why worry about whether a signal is a power or an energy signal? The answer is because the formulas for convolution, correlation, and Fourier transforms are different for each signal class.

Example 1.3. The square pulse in Fig. 1.9 has zero power. To see this, apply Eq. 1.2 with increasing values of a as follows:

$$\text{for } a = 5: \quad P = \frac{1}{10} \int_0^1 2^2 dt = \frac{4}{10}$$

$$\text{for } a = 10: \quad P = \frac{1}{20} \int_0^1 2^2 dt = \frac{4}{10}$$

$$\text{for } a = 100: \quad P = \frac{1}{200} \int_0^1 2^2 dt = \frac{4}{200}$$

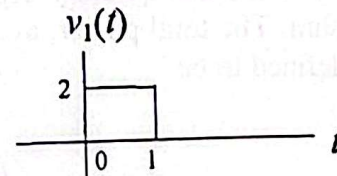


Figure 1.9. Square Pulse.

As a gets larger and larger, the power gets smaller and smaller, until in the limit it is zero. The energy in the pulse is finite, however, because Eq. 1.3 gives

$$E = \int_0^1 2^2 dt = 4$$

Since the power is zero and the energy is finite, this is an energy signal. ■

Example 1.4. The periodic square wave in Fig. 1.10 has finite power and infinite energy. Applying Eq. 1.5 gradually, as we did above, gives

$$\text{for } a = 5 \quad P = \frac{1}{10} \int_{-5}^5 v_2^2(t) dt = \frac{20}{10}$$

$$\text{for } a = 10 \quad P = \frac{1}{20} \int_{-10}^{10} v_2^2(t) dt = \frac{40}{20}$$

$$\text{for } a = 100 \quad P = \frac{1}{200} \int_{-100}^{100} v_2^2(t) dt = \frac{400}{200}$$

Larger and larger values of a result in the same power, 2. Notice that power in a periodic signal of period T is given by the simplified formula

$$P = \frac{1}{T} \int_0^T v^2(t) dt \quad (1.4)$$

which should be familiar from circuit analysis courses. The rms value is the square root of the power. Therefore, the rms value of the signal in this example is $\sqrt{2}$. The energy in the signal is infinite because

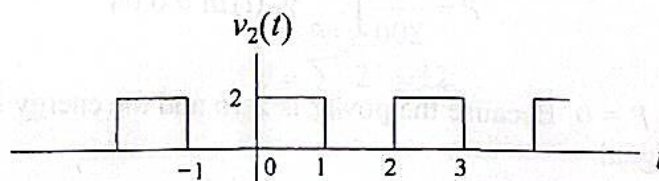


Figure 1.10. Periodic square wave with finite power and infinite energy.

$$E = \int_{-\infty}^{\infty} v_2^2(t) dt = \infty$$

Since the energy is infinite and the power is finite, this is a power signal. ■

Any signal that is actually generated and used must be of finite duration, and therefore an energy signal. The classification above applies only to our mathematical model of the signal. We will often say that a signal lasts forever, although this is practically impossible. However, just because a signal lasts forever does not mean that it is a power signal. Here is an example.

Example 1.5. The decaying exponential in Fig. 1.11 lasts forever but has finite energy.

$$E = \int_{-\infty}^{\infty} v_3^2(t) dt = 2 \int_0^{\infty} 4e^{-2t} dt = 4$$

The power is zero, as indicated in the following sequence of integrals.

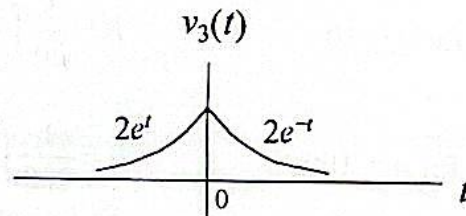


Figure 1.11. Decaying exponential with finite energy.

for $a = 5$:
$$P = \frac{1}{10} \int_{-5}^5 v_3^2(t) dt = 0.4$$

for $a = 10$:
$$P = \frac{1}{20} \int_{-10}^{10} v_3^2(t) dt = 0.2$$

for $a = 100$:
$$P = \frac{1}{200} \int_{-100}^{100} v_3^2(t) dt = 0.04$$

In the limit, $P = 0$. Because the power is zero and the energy is finite, this is an energy signal. ■

Drill 1.1. Find the power and energy in $v(t) = e^{-2t}u(t)$.

Answer: $P = 0$, $E = 1/4$.

For discrete-time signals these definitions become

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N v^2(n) \quad (1.5)$$

and

$$E = \sum_{n=-\infty}^{\infty} v^2(n) \quad (1.6)$$

In Eq. 1.5, the sum over the range $(-N, N)$ has $2N+1$ terms in it, so divide by this number of terms to find the average. This is also called the *mean-square value*. The mean-square value is *average power*, although the concept of power loses its meaning for discrete-time signals. The energy in Eq. 1.6 is the *total energy*, not the instantaneous energy.

Example 1.6. The signal $x(n)$ shown in Fig. 1.12 is an energy signal. To see this, apply Eq. 1.5 with increasing values of N to get

for $N = 5$:
$$P = \frac{1}{11} \sum_{n=-5}^5 x^2(n) = \frac{12}{11}$$

for $N = 10$:
$$P = \frac{1}{21} \sum_{n=-10}^{10} x^2(n) = \frac{12}{21}$$

for $N = 100$:
$$P = \frac{1}{201} \sum_{n=-100}^{100} x^2(n) = \frac{12}{201}$$

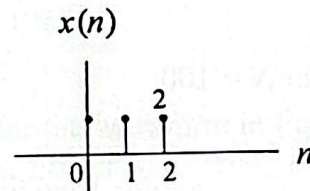


Figure 1.12. Energy signal.

As N gets larger and larger, the power gets smaller and smaller, until in the limit it is zero. The energy in the pulse is finite, however, because Eq. 1.9 gives

$$E = \sum_{n=0}^2 2^2 = 12$$

Since the power is zero and the energy is finite, this is an energy signal. ■

Example 1.7 The periodic digital square wave in Fig. 1.13 has finite power and infinite energy. Apply Eq. 1.5 gradually, as above, to get

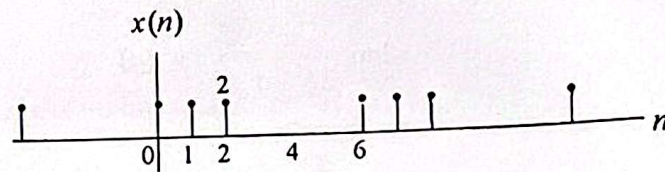


Fig. 1.13. A periodic digital signal with finite power and infinite energy.

for $N = 5$:

$$P = \frac{1}{11} \sum_{n=-5}^5 x^2(n) = \frac{20}{11}$$

for $N = 10$:

$$P = \frac{1}{21} \sum_{n=-10}^{10} x^2(n) = \frac{40}{21}$$

for $N = 100$:

$$P = \frac{1}{201} \sum_{n=-100}^{100} x^2(n) = \frac{400}{201}$$

As this continues with larger and larger values of N , P approaches the value 2. Notice that here, too, the power in a periodic signal with period N can be found by averaging over only one period.

$$P = \frac{1}{N} \sum_{n=0}^{N-1} x^2(n) \quad (1.7)$$

The rms value is again the square root of the power. The energy is given by

$$E = \sum_{n=-\infty}^{\infty} x^2(n) = \infty$$

Since the energy is infinite and the power is finite, this is a power signal. ■

As with continuous time signals, just because a signal lasts forever does not mean that it is a power signal. Here is an example to illustrate that fact for discrete-time signals.

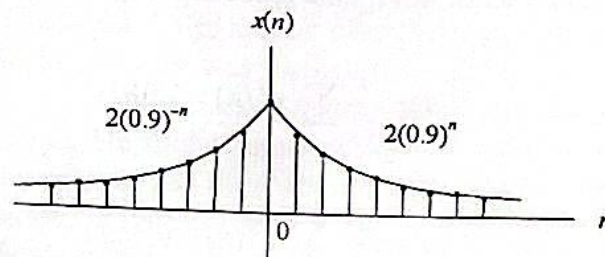


Figure 1.14. Exponential energy signal.

Example 1.8 The exponential signal in Fig. 1.14 lasts forever in both directions, but it is an energy signal. The energy is

$$E = \sum_{n=-\infty}^{\infty} x^2(n) = 2 \sum_{n=1}^{\infty} x^2(n) + x^2(0)$$

The terms on the right are derived by noticing that for the waveform in Fig. 1.14, the sum from $-\infty$ to -1 is the same as the sum from 1 to ∞ . Then the energy is given by twice the sum from 1 to ∞ plus the term for $n = 0$. The

sum $\sum_{n=0}^{\infty} (0.81)^n$ is a geometric series with value given by

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad |a| < 1 \quad (1.8)$$

The sum in the expression for energy has a lower limit of 1 instead of 0 , so calculate the sum using Eq. 1.8 and then subtract the $n = 0$ term. With $a = 0.81$ this gives

$$\sum_{n=1}^{\infty} x^2(n) = 4 \sum_{n=0}^{\infty} (0.81)^n - 4 = \frac{4(0.81)}{1-0.81} = 17.0526$$

Therefore, the energy is double this number plus $x^2(0)$.

$$E = 2(17.0526) + 4 = 38.1053$$

The power is zero, as shown by the following sequence of sums.

for $N = 5$:

$$P = \frac{1}{11} \sum_{n=-5}^5 x^2(n) = 2.383$$

for $N = 10$:

$$P = \frac{1}{21} \sum_{n=-10}^{10} x^2(n) = 0.5471$$

for $N = 100$:

$$P = \frac{1}{201} \sum_{n=-100}^{100} x^2(n) = 0.0552$$

In the limit, $P = 0$. Because the power is zero and the energy is greater than zero but finite, this is an energy signal. ■

We will use only those signals that can be classified into one of the two categories of energy signal or power signal, according to the value of P or E from these definitions. Notice once again that a power signal must last forever, so these definitions apply only to their mathematical model.

Drill 1.2. Find the power and energy in $v(n) = (0.9)^n u(n)$.

Answer: $P = 0$, $E = 5.2632$.

1.4. Deterministic and Random Signals

Different definitions for a random signal exist in the literature, which means that there is no good definition. *Random* means unpredictable, hence a random signal is one with unpredictable values. Wavelet and Fourier transforms apply to random signals as well as to deterministic signals. For now, let us review some history that should help you better understand random noise.

The Scottish botanist Robert Brown (1773-1858) was among the leading scientists of his day. He met with and advised the young Charles Darwin (1809-1882) before Darwin's famous voyage on the *Beagle* beginning in 1831. Brown himself had served as the naturalist on a voyage to the still-new and largely unexplored continent of Australia in 1801. His ship returned in 1805 with some 4,000 species of plants.

Brown is remembered particularly for two discoveries. He recognized the small body within cells as a regular feature, and in 1831 gave it the name by which it has been known ever since: *nucleus*, from the Latin word meaning "little nut." The second discovery had repercussions outside the life sciences. In 1827, he was viewing wheat pollen suspended in water under a microscope when he noticed that the particles were moving about in an irregular fashion. He studied other particles suspended in water, even a particle from the sphinx, and observed this same random motion. This has been called *Brownian motion* ever since. Brown believed that he had discovered a fundamental form of life that was "present in animate and inanimate objects," and no one proved otherwise during his lifetime.

Ludwig Boltzmann's tombstone in the central cemetery of Vienna has on it his name, the dates 1844-1906, and the inscription $S = k \log W$. In this formula, S stands for entropy, k is now known as Boltzmann's constant, and W is a measure of the possible states of nature. This is one of the most remarkable formulas in science, for it quantifies the chaos that exists at the atomic level in matter and applies to any system with random parameters. Boltzmann was an early proponent of atomism at a time when there was considerable debate about the nature of matter. His work influenced the work of Einstein on Brownian motion.

In 1905, Albert Einstein (1879-1955) published five papers involving three developments of major importance in the German journal *Annalen der Physik* (Annals of Physics). His theory of relativity is so startling and contrary to common experience that we often ignore the other two developments. His first paper dealt with the photoelectric effect, and his second paper worked out a mathematical analysis of Brownian motion. Einstein showed that if the water in which the particles were suspended was composed of molecules in random motion according to the kinetic theory of Boltzmann, the result would be the observed Brownian motion of the suspended particles. Einstein went on to say that any system in thermal equilibrium would exhibit this Brownian motion, and he correctly derived the current that flows in a circuit due to the random motion of electrons. Since an electric circuit element such as a resistor is a system in thermal equilibrium, random noise will be present as a voltage across its terminals. Einstein's formula gives the value of this voltage as

$$V = 4kTR \times Bw$$

where V is the rms voltage, k is Boltzmann's constant, T is the temperature in Kelvin, R is the value of the resistance in ohms, and Bw is the bandwidth of the measuring instrument. For example, a 1-k Ω resistor at room

temperature, measured by an ordinary voltmeter, has an rms voltage across its terminals of 3 to 6 μV . The waveform in Fig. 1.15 shows a typical oscilloscope image of this voltage.

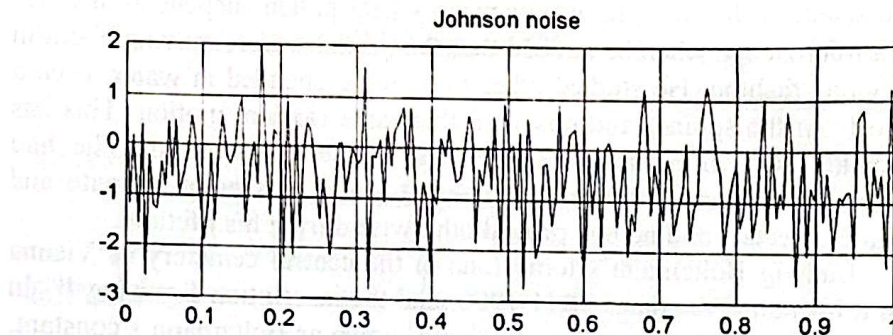


Figure 1.15. Typical Johnson noise.

This noise, called *Johnson noise* after the scientist who first measured it at Bell Labs in 1928, limits communication. A distant transmitter may be so weak that this noise obliterates the signal. One way to combat this limitation is to reduce the temperature in the front end of the receiver. Thus, some sensitive receivers operate in a cold environment, with the components in the first few stages of amplification reduced in temperature to near absolute zero.

1.5. Fourier and Haar Transforms

Signals fall naturally into one of four categories:

1. Continuous-time energy signals.
2. Continuous-time power signals.
3. Discrete-time energy signals.
4. Discrete-time power signals.

There is one form of the Fourier transform for each category

1. Continuous-time Fourier transform (CTFT).
2. Continuous-time Fourier series (CTFS).
3. Discrete-time Fourier transform (DTFT).
4. Discrete-time Fourier series (DTFS).

The matchup goes like this:

1. Continuous-time energy signals. $\leftrightarrow$ CTFT.
2. Continuous-time power signals. $\leftrightarrow$ CTFS.
3. Discrete-time energy signals. $\leftrightarrow$ DTFT.
4. Discrete-time power signals. $\leftrightarrow$ DTFS.

The fast Fourier transform is a fast version of the DTFS and does not apply to any of the other transforms. The DTFS is defined by

$$V(k) = \sum_{n=0}^{N-1} v(n) e^{-j2\pi nk/N} \quad (1.9)$$

This is the only one of the four Fourier transforms that will be of much concern to us in this book.

Note that Eq. 1.9 is really N equations, one for each value of k .

$$\begin{aligned} V(0) &= \sum_{n=0}^{N-1} v(n) e^0 = \sum_{n=0}^{N-1} v(n) \\ V(1) &= \sum_{n=0}^{N-1} v(n) e^{-j2\pi n/N} \\ &\vdots \\ V(N-1) &= \sum_{n=0}^{N-1} v(n) e^{-j2\pi n(N-1)/N} \end{aligned}$$

There is a different exponential term used in each equation, starting with e^0 , $e^{-j2\pi n/N}$, and progressing to $e^{-j2\pi n(N-1)/N}$. These are the basis functions for the expansion. (Chapter 3 covers the concept of basis functions in detail.) In other words, the DTFS is an expansion of $v(n)$ in terms of these exponentials. These exponential terms are usually written in a standard form. Let

$$W_N = e^{-j2\pi/N} \quad (1.10)$$

Then

$$W_N^0 = e^0$$

$$W_N^1 = e^{-j2\pi/N}$$

$$W_N^2 = e^{-j2\pi 2/N}$$

$$\vdots$$

$$W_N^{N-1} = e^{-j2\pi(N-1)/N}$$

These N equations can be put in matrix form. For example, if $N = 4$, the matrix is given by

$$\begin{bmatrix} V(0) \\ V(1) \\ V(2) \\ V(3) \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} v(0) \\ v(1) \\ v(2) \\ v(3) \end{bmatrix} \quad (1.11)$$

Using this notation, Eq. 1.9 can be written as

$$V(k) = \sum_{n=0}^{N-1} v(n) W_N^n \quad (1.12)$$

Example 1.9. Find the DTFS for $v(n) = \{1, 2, 3, 4\}$.

Solution: Here, $N = 4$, so we can use Eq. 1.11 directly. Multiplying by the matrix in Eq. 1.11 gives

$$\begin{bmatrix} V(0) \\ V(1) \\ V(2) \\ V(3) \end{bmatrix} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} v(0) \\ v(1) \\ v(2) \\ v(3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ -2 + j2 \\ -2 \\ -2 - j2 \end{bmatrix}$$

Haar Transform

The Haar transform is one (of several) wavelet transforms that can be calculated with a formula. The formula (actually, two formulas) is given by

$$c_{00} = \int_0^1 v(t) \varphi_{00}(t) dt \quad (1.13a)$$

$$d_{kj} = \int_0^1 v(t) \psi_{kj}(t) dt \quad (1.13b)$$

where the basis functions φ_{00} and ψ_{kj} are as shown in Fig. 1.16. The function $\varphi_{00}(t)$ is called the *scaling function*, and the $\psi_{kj}(t)$ are called *wavelets*.

This diagram shows the level 0, 1, and 2 Haar basis functions, but these can continue to higher and higher levels. Level 0 contains the functions $\varphi_{00}(t)$ (called the *scaling function*) and $\psi_{00}(t)$ (called the *mother wavelet*). Level 1 contains the two functions $\psi_{10}(t)$ and $\psi_{11}(t)$. Level 2 contains the four functions $\psi_{20}(t)$, $\psi_{21}(t)$, $\psi_{22}(t)$, and $\psi_{23}(t)$. Each higher level contains twice as many wavelets $\{\psi_{jk}\}$. Thus, level 3 contains eight wavelets, each just half as long as the level 2 wavelets and scaled appropriately to have unit energy. To calculate the Haar transform, expand the time function in terms of these basis functions. Equation 1.13 gives the formulas for calculating the transform.

Example 1.10. Calculate the Haar transform of the signal in Fig. 1.17.

Solution: Applying Eq. 1.13 to the function $v(t)$ gives the following numbers.

$$c_{00} = \int_0^1 v(t) \varphi_{00}(t) dt = 0$$

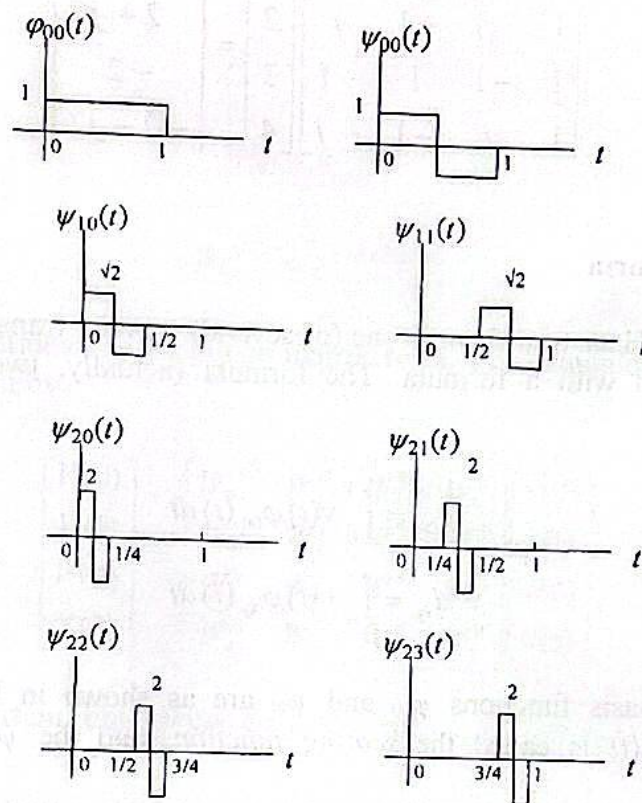
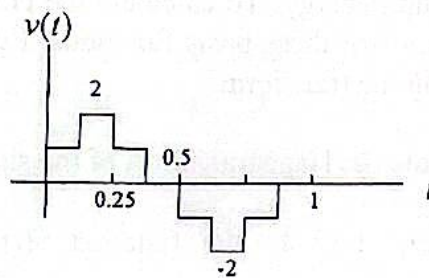


Figure 1.16. Haar basis functions to level 2.

Figure 1.17. The function $v(t)$.

$$d_{00} = \int_0^1 v(t) \psi_{00}(t) dt = 1$$

$$d_{10} = \int_0^{0.5} v(t) \psi_{10}(t) dt = \frac{\sqrt{2}}{4}$$

$$d_{11} = \int_{0.5}^1 v(t) \psi_{11}(t) dt = -\frac{\sqrt{2}}{4}$$

$$d_{20} = \int_0^{0.25} v(t) \psi_{20}(t) dt = -\frac{1}{4}$$

$$d_{21} = \int_{0.25}^{0.5} v(t) \psi_{21}(t) dt = \frac{1}{4}$$

$$d_{22} = \int_{0.5}^{0.75} v(t) \psi_{22}(t) dt = \frac{1}{4}$$

$$d_{23} = \int_{0.75}^1 v(t) \psi_{23}(t) dt = -\frac{1}{4}$$

Thus, the first eight terms of the Haar transform of $v(t)$ consist of the eight values $\left\{0, 1, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}\right\}$.

Equation 8.12 gives the equation for the inverse wavelet transform. When applied to the function in Fig. 1.17, it takes the form

$$v(t) = c_{00} \varphi_{00}(t) + d_{00} \psi_{00}(t) + d_{10} \psi_{10}(t) + \cdots + d_{23} \psi_{23}(t)$$

Example 1.11. Show that the inverse transform gives the function $v(t)$ in Fig. 1.17.

Solution: Multiplying the functions in Fig. 1.16 by the coefficients, $c_{00} = 0$, $d_{00} = 1$, and so on, results in the functions in Fig. 1.18. If these functions are summed, the result is $v(t)$ in Fig. 1.17. (Try it if you don't believe it.)

From these examples one can see many similarities between Fourier and wavelet transforms. Time functions are expanded in terms of basis functions in both transforms. These basis functions are exponentials for the Fourier transform, and they are square pulses for the Haar transform. The coefficients are calculated by an inner product operation [i.e., by integrating the product of $v(t)$ and basis function].

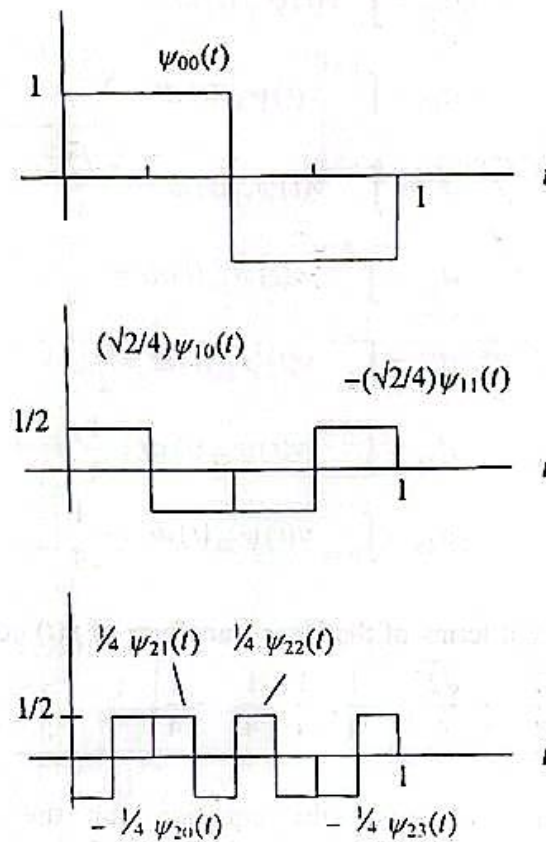


Figure 1.18. Sum these functions to obtain $v(t)$.

However, there are differences that are not apparent in these simple examples. There are many wavelet basis functions other than the Haar functions, and there is one wavelet transform for each set of basis functions. This is similar to the Fourier transforms, where there are four forms of the Fourier transform, but not similar in that there is a vast array of wavelet basis functions. Another important difference is in the way the coefficients for the wavelet transform are calculated. Note that in the above example using an inner product formulation, the formula for the basis function must be known. Most wavelet coefficients are calculated in a different way, using multirate sampling theory. There it is necessary to know only the filter coefficients. This method does not use the basis functions. We will devote much of this book to this procedure